Problem J. Best Carry Player 3

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	1024 megabytes

After learning the bitwise XOR operation, Little Cyan Fish would like to practice it by playing the following game.

Little Cyan Fish has an integer X. He wants to transform X into another integer Y using addition, subtraction, or bitwise XOR operations.

Because he is not a *Best Carry Player*, he cannot understand the operations of addition and subtraction between large numbers. Thus, he can only perform the following operations:

- $(+)$: Change X to $X + 1$.
- $(-)$: Change X to $X - 1$. This operation is not available if $X = 0$.
- $(\oplus)$: Choose an integer $0 \le t \le K$, change X to $X \oplus t$, where $\oplus$ is the bitwise XOR operator.

Given integers X, Y and K, you need to calculate the minimum number of operations that Little Cyan Fish needs to perform to transform X into Y.

Input

There are multiple test cases. The first line contains one integer T ($1 \le T \le 10^5$), representing the number of test cases.

For each test case, the first line contains three integers X, Y and K ($0 \le X, Y, K < 2^{60}$).

Output

For each test case, output a single line contains a single integer, indicating the minimum number of operations that Little Cyan Fish needs to perform.

Example

standard input	standard output
8 4 5 0 5 8 3 9 2 6 15 28 5 97 47 8 164 275 38 114514 1919 810 0 1152921504606846975 1	1 2 3 5 11 6 331 1152921504606846975

Note

In the first test case, the optimal plan is:

- $(+)$: Change X from 4 to $4 + 1 = 5$.

So the answer is 1.

In the second test case, the optimal plan is:

- ($\oplus 2$): Choose $t = 2$, change X from 5 to $5 \oplus 2 = 7$.
- ($+$): Change X from 7 to $7 + 1 = 8$.

So the answer is 2.

In the third test case, the optimal plan is:

- ($-$): Choose X from 9 to $9 - 1 = 8$.
- ($-$): Change X from 8 to $8 - 1 = 7$.
- ($\oplus 5$): Choose $t = 5$, Change X from 7 to $7 \oplus 5 = 2$.

So the answer is 3.